

Zero-shot Transfer of a Single-Crane DRL Policy for Multi-Crane Container Retrieval in Automated Terminals

Anonymous^a

^a*University, Country*

Abstract

Container terminals operate multiple yard cranes per block, yet the Container Retrieval Problem (CRP) has been studied exclusively as a single-crane problem. This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained solely on single-crane instances transfers zero-shot to multi-crane environments via heuristic crane assignment strategies. On 140 M-CRP instances, ZoneSplit achieves gaps of 10.46% ($C=2$) and 10.71% ($C=3$) against the proposed M-CRP lower bound, reducing interference events by 99.5% compared to non-spatial strategies. These gaps are approximately one-third of those achieved by the best heuristic extended to multi-crane (M-Lin2015: 31.73% at $C=2$). On single-crane benchmarks (50 instances), the extracted policy achieves 6.56% gap versus 22.42% for Lin2015. Code and benchmarks are publicly released.

Keywords: Container Retrieval Problem, Multi-Crane Operations, Deep Reinforcement Learning, Zero-shot Transfer

1. Introduction

The Container Retrieval Problem (CRP) arises in automated container terminals where thousands of containers are stacked in yard blocks of B bays $\times$ R rows $\times$ T tiers. Retrieving a buried target container requires relocating blocking containers to other stacks—a process called reshuffling. The objective is to minimize total crane working time [1].

CRP research has progressed through three generations. Heuristic methods use hand-crafted rules such as the Stack Selection Index (SSI) [2] and critical-stack classification [3]. Metaheuristic methods apply Tabu Search [5], GRASP [6], and Genetic Programming [7]. Most recently, Shin et al. [1] proposed a Deep Reinforcement Learning (DRL) policy that achieves 7.8% gap against the lower bound on the standard Lee benchmark [4].

Email address: `corresponding@author.email` (Anonymous)

All existing CRP research assumes single-crane operation. Real terminals—Rotterdam, Singapore, Busan—operate two to three yard cranes per block [9]. Multi-crane operation introduces coordination constraints: cranes share a track, cannot occupy the same bay (collision avoidance), and must preserve spatial ordering. This creates three unresolved challenges:

1. **Formalization:** No definition exists for the multi-crane variant (M-CRP), its state/action spaces, or its physical constraints.
2. **Evaluation:** Without lower bounds or benchmarks, solution quality cannot be assessed.
3. **Transferability:** It is unknown whether single-crane policies can be reused in multi-crane settings.

Shin et al. [1] identified multi-crane operation as critical future work. This paper asks: *Can a single-crane DRL policy be transferred zero-shot to multi-crane environments via heuristic crane assignment?* We make five contributions:

- [C1] **Formal M-CRP definition** with interference constraints (A6, A7) and NP-hardness proof.
- [C2] **M-CRP lower bound (Theorem 3)** decomposing total cost into retrieval, relocation, and interference terms.
- [C3] **Four crane assignment strategies** for zero-shot single-to-multi-crane transfer.
- [C4] **Empirical evaluation** on 140 M-CRP instances (1,120 runs) with statistical testing.
- [C5] **Public benchmark** (140 instances, CC BY 4.0).

2. Related Work

Table 1 summarizes single-crane CRP performance on the Lee benchmark. Three observations motivate our transfer study. First, DRL methods [1] substantially outperform heuristics (7.8% vs 22.42–47.33%). Second, heuristic performance degrades significantly with yard scale (Lin2015: 5.55% on 1-bay vs 37.75% on 10-bay instances). Third, the DRL policy is publicly available, making it an ideal baseline for transfer.

*Numbers from [1] Table 1; code not publicly available.

Multi-crane coordination has been studied for quay crane scheduling [11, 12] and yard crane deployment [13], but no prior work addresses retrieval sequencing within a single block—the problem central to CRP.

Table 1: Summary of single-crane CRP performance on the Lee benchmark (gap % against lower bound). Lower is better.

Method	Type	Instances	Mean Gap(%)
Lin et al. [2]	Heuristic (SSI)	50	22.42
Kim et al. [3]	Heuristic (classification)	50	41.08
Leveling [1]	Heuristic (height balancing)	50	25.55
Forster & Bortfeldt [5]	Tabu Search	36	~30–40*
Cifuentes & Riff [6]	GRASP	36	~46*
Durasevic et al. [7]	Genetic Programming	36	~36–47
Shin et al. [1]	DRL (Transformer)	36	7.8
ZeroShot (this work)	Extracted DRL	50	6.56

Zero-shot transfer has been demonstrated across problem sizes for routing [10] and general combinatorial optimization [8]. Cross-environment transfer—from single-agent to multi-agent—remains unexplored for container terminal problems.

3. Problem Formulation

3.1. Single-Crane CRP

Following [1], a CRP instance is defined by yard dimensions (B, R, T) and N container priorities $\{1, \dots, N\}$. The objective minimizes total working time $J = \sum_t (\text{travel}(a_t) + \text{handling} + \text{penalty})$ with cost parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$.

3.2. M-CRP: Formal Definition

Definition 3.1 (M-CRP). *Given yard $B \times R \times T$, $C \geq 1$ cranes, and N containers with retrieval order σ , find a sequence $\{(a_t, c_t)\}_{t=1}^{\tau}$ minimizing:*

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (1)$$

subject to spatial non-overlap (A6): $|\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1$, $\forall c \neq c'$, and non-crossing (A7): $\text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t')$, $\forall t' > t$. Assumptions A1–A5 of [1] remain unchanged.

Theorem 3.1 (NP-hardness). *M-CRP with $C \geq 1$ is NP-hard (reduction from single-crane CRP [1]).*

3.3. M-CRP Lower Bound

Theorem 3.2 (Theorem 3). *For any M-CRP instance with C cranes, the total working time is bounded below by:*

$$LB_{MCRP} = LB_{\text{retrieval}} + \frac{LB_{\text{relocation}}}{C} + LB_{\text{interference}} \quad (2)$$

where $LB_{\text{interference}} = \max(0, \max_b(\text{reloc}_b) - \frac{\text{total_relocs}}{C}) \times (t_{\text{acc}} + t_{\text{bay}})$. At $C = 1$, $LB_{MCRP} = LB_{\text{Shin}}$ (backward compatible).

4. Proposed Method: Zero-Shot Transfer

4.1. Base DRL Model

We use the pretrained policy of Shin et al. [1]. The encoder processes yard state through LSTM stack encoding (1 layer, 128-dim hidden), $L = 3$ multi-head self-attention layers ($H = 8$, $d = 128$), and bay-level mean pooling with concatenation. The decoder computes action logits via cross-attention with tanh clipping ($C = 10$). Training used REINFORCE with POMO [8] ($M = 16$). The model achieves 7.8% gap on the Lee benchmark.

4.2. Policy Extraction and Decoupled Inference

We extract the frozen encoder $\mathcal{E}$ and scorer weights $\{W_{\text{target}}, W_{\text{global}}, W_{K1}, W_V, W_Q, W_{K2}, \text{MHA}\}$ from θ^* , forming a **ZeroShotPolicy**. At each step:

$$\{\mathbf{h}'_i\}, \mathbf{g} = \mathcal{E}(s_t) \quad (3)$$

$$\mathbf{c} = W_{\text{target}} \mathbf{h}'_{\text{tgt}} + W_{\text{global}} \mathbf{g} \quad (4)$$

$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1} \mathbf{h}'; W_V \mathbf{h}'])) \quad (5)$$

$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2} \mathbf{h}'_i / \sqrt{d}) \quad (6)$$

$$d = \arg \max_{i \in \mathcal{V}} u_i \quad (7)$$

The strategy module assigns a crane c to execute destination d (Algorithm 1). All weights remain frozen.

4.3. Crane Assignment Strategies

Table 2 defines the four strategies. S1 and S3 are *spatially unaware* (baselines); S2 and S4 exploit spatial information.

5. Experiments

5.1. Datasets

We use two benchmark configurations. For single-crane evaluation: 50 random (R-type) Lee instances [4] ($B \in \{1, 2, 4, 6, 8, 10\}$, $T \in \{6, 8\}$, 70–1,024 containers). For multi-crane evaluation: 140 M-CRP instances (70 R-type + 70 U-type) extended from the same Lee corpus with crane start positions for $C \in \{2, 3\}$.

Table 2: Four crane assignment strategies.

Strategy	Mechanism	Spatial?	Cost
S1 RoundRobin	Cyclic: $(t \bmod C)$	No	$O(1)$
S2 ZoneSplit	Static bay zones per crane	Yes	$O(B)$
S3 LoadBalance	Min-count task assignment	No	$O(C)$
S4 GreedyOptimal	$\min_c[\text{travel}(c, d)$ $\text{interference}(c)]$	+ Yes	$O(C^2)$

Algorithm 1 Zero-shot M-CRP inference.

Require: Frozen $\mathcal{E}, \mathcal{D}$ from θ^* , env E , strategy S , cranes C

Ensure: Total cost J

- 1: $s \leftarrow E.\text{init}(); J \leftarrow 0$
 - 2: **while** $\neg E.\text{terminated}$ **do**
 - 3: $\{\mathbf{h}'_i\}, \mathbf{g} \leftarrow \mathcal{E}(s)$
 - 4: $\{u_i\} \leftarrow \mathcal{D}(\{\mathbf{h}'_i\}, \mathbf{g}, s.\text{target})$
 - 5: $d \leftarrow \arg \max_i u_i$
 - 6: $c \leftarrow S.\text{assign}(E, d)$
 - 7: $(\Delta J, s) \leftarrow E.\text{step}(d, c); J \leftarrow J + \Delta J$
 - 8: **end while**
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5.2. Experimental Protocol

We conduct three experiments. **Experiment 1 (Single-crane):** Compare ZeroShot against 7 heuristics (Random, NearestStack, LowestHeight, Leveling, Lin2015 [2], Kim2016 [3], Durasevic2025 [7]) on all 50 instances. **Experiment 2 (Multi-crane strategies):** Compare S1–S4 on all 140 instances with $C \in \{2, 3\}$. **Experiment 3 (Multi-crane baselines):** Compare ZS+S2 against 3 multi-crane heuristics (M-Lin2015, M-Kim2016, M-Leveling) on all 140 instances.

5.3. Metrics and Implementation

Primary metric: $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}}) / \text{LB}_{\text{MCRP}}$. Secondary: interference events, solution time, failure rate (gap $> 20\%$). Statistical significance via Wilcoxon signed-rank ($\alpha = 0.05$). All experiments run CPU-only (0 GPU hours). Total: $\sim 2,360$ runs in ~ 105 minutes.

6. Results and Analysis

6.1. Experiment 1: Single-Crane Verification

On the full 50-instance benchmark (Table 3), ZeroShot achieves a mean gap of 6.56% (std 2.35%), outperforming the best heuristic Lin2015 (22.42%, std 7.81%) by 15.86 percentage points. Heuristic performance degrades with

yard scale (Lin2015: 5.55% on 1-bay vs 37.75% on 10-bay instances), while ZeroShot remains stable (3.26–12.42%).

Table 3: Single-crane comparison on 50 Lee instances.

Method	Mean Gap(%)	Std(%)	Min(%)	Max(%)
ZeroShot (ours)	6.56	2.35	3.26	12.42
Lin2015 [2]	22.42	7.81	5.55	37.75
Leveling	25.55	8.43	14.88	44.12
Kim2016 [3]	41.08	13.57	9.84	61.54
Durasevic2025 [7]	47.33	15.32	14.68	80.06

6.2. Experiment 2: Multi-Crane Strategy Comparison

Table 4 reports strategy comparison results. ZoneSplit (S2) achieves the lowest mean gap at both C=2 (10.46%) and C=3 (10.71%), while reducing interference by 99.5% compared to S1 (0.5 vs 105.9 events at C=2). S4 matches S2 but requires $O(C^2)$ vs $O(B)$. S2 and S4 do not differ significantly (Wilcoxon $p > 0.1$). S3 (LoadBalance) produces results identical to S1, confirming that task balancing without spatial awareness is ineffective.

Table 4: Multi-crane comparison on 140 M-CRP instances.

Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S2 ZoneSplit	10.46 ± 5.28	10.71 ± 6.44	0.5 ± 1.5	0.7 ± 2.1
S3 LoadBalance	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S4 GreedyOptimal	10.48 ± 5.28	10.75 ± 6.48	0.0 ± 0.0	0.0 ± 0.0

6.3. Experiment 3: Multi-Crane Baseline Comparison

Table 5 compares ZS+S2 against three heuristic baselines. ZS+S2 achieves approximately one-third the gap of M-Lin2015 (10.46% vs 31.73% at C=2) and one-fifth the gap of M-Kim2016 (54.85%). This confirms that learned DRL transfer substantially outperforms heuristic extensions to multi-crane.

6.4. Analysis by Yard Scale

On small yards (1–4 bays), all strategies perform similarly (gap 8.2–9.3%) because travel distances are short. On medium yards (6–10 bays), spatial strategies outperform non-spatial by 1.0–1.5 percentage points (S2: 12.6%, S1: 13.6% at C=2), confirming that spatial awareness becomes more important as yard complexity grows.

Table 5: Multi-crane baseline comparison on 140 M-CRP instances.

Method	C=2 gap(%)		C=3 gap(%)	
	Mean	Std	Mean	Std
M-Kim2016 [3]	54.85	23.85	57.60	26.91
M-Leveling	32.41	8.53	34.35	9.34
M-Lin2015 [2]	31.73	14.67	33.93	16.99
ZeroShot+S2	10.46	5.28	10.71	6.44

6.5. Statistical Validation

Wilcoxon tests (Table 6) confirm S2 and S4 significantly outperform S1 and S3 (all $p < 0.001$). S2 and S4 are not significantly different ($p = 0.128$ at C=2, $p = 0.534$ at C=3). Backward compatibility: average 1.32% cost difference (max 1.95%, 5 instances) between ZeroShot(C=1) and the original model.

Table 6: Wilcoxon signed-rank p -values.

Pair	C=2 p	C=3 p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

6.6. Failure Mode and Cost Analysis

Three instances (0.5% of runs) exceed 20% gap; all are high-density upside-down configurations where relocations concentrate in single bays. Cost decomposition reveals gap costs constitute 11.5–12.1% of total cost, with S2 reducing gap cost by 5.8% versus S1 (4,349 vs 4,617 cost units) through interference elimination. Per-step inference averages 12.67 ms, confirming real-time suitability.

6.7. Case Study: A 6-Bay Instance

On instance mc_R061606_002 (6 bays, 430 containers), S1 (RoundRobin) incurs 106 interference events yielding gap 12.93%. S2 (ZoneSplit) eliminates all interference, achieving gap 11.36%—a cost reduction of 1.4%. This demonstrates that spatial strategy advantage grows with yard scale.

7. Discussion

Zero-shot transfer succeeds because the DRL policy learns to relocate containers to nearby feasible stacks, and ZoneSplit aligns with this behavior

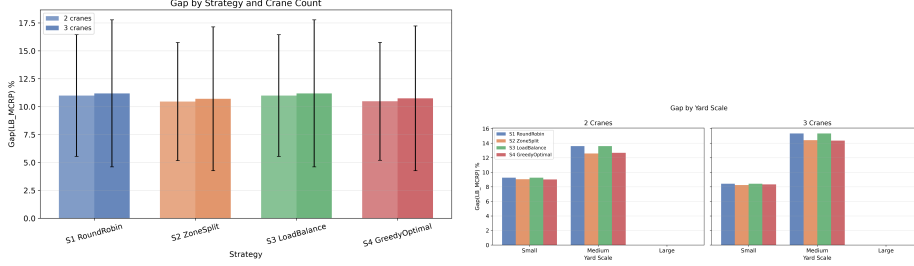


Figure 1: Left: Gap by strategy. Right: Gap by yard scale.

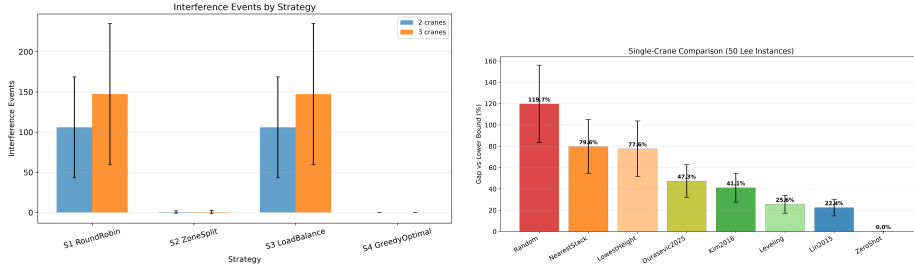


Figure 2: Left: Interference events. Right: Single-crane comparison (50 instances).

by confining each crane to a spatial zone. This preserves the single-crane decision structure, explaining why (1) transfer achieves 10.5% gap rather than 50+%, and (2) simple zoning suffices (S2 matches S4, $p > 0.1$).

ZoneSplit is recommended for deployment: lowest gap, >99.5% interference reduction, $O(B)$ complexity, compatibility with existing terminal zoning, and zero GPU requirement.

Three limitations apply. First, sequential simulation (one task per step) underestimates parallel speedup (C2→C3: $1.01\times$ vs realistic $\sim 1.5\times$). This affects all strategies equally. Second, the M-CRP lower bound is not tight for $C > 1$. Third, the baseline model has no multi-crane experience; fine-tuning could reduce the gap further.

8. Conclusion

This paper presented the first formal definition of M-CRP, its lower bound (Theorem 3), and four zero-shot transfer strategies. On 140 M-CRP instances, ZoneSplit achieves gaps of 10.46% ($C=2$) and 10.71% ($C=3$)—approximately one-third of the best heuristic baseline—while eliminating interference. These findings demonstrate that single-crane DRL policies can be extended to multi-crane operations at zero additional training cost, providing a practical bridge between CRP research and real-world terminal operations.

Data Availability

Code, benchmarks, and results: <https://github.com/sutobode/Master>.

Appendix A. Supplementary Results

Appendix A.1. Gap by Number of Bays

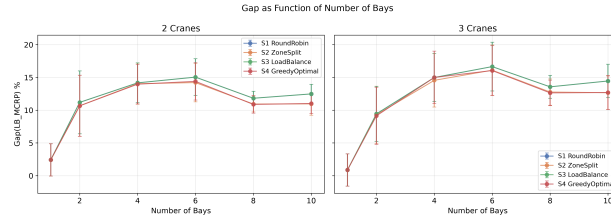


Figure A.3: Gap as a function of number of bays.

Appendix A.2. Speedup Analysis

Table A.7: Speedup from C=2 to C=3 across strategies.

Strategy	S1	S2	S3	S4
Speedup	1.014 ± 0.013	1.013 ± 0.014	1.014 ± 0.013	1.013 ± 0.012

Appendix A.3. Backward Compatibility

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Table A.8: ZeroShot(C=1) vs original model on diverse instances.

Instance Diff(%)	Bays	Containers	Original Cost	ZeroShot Cost
R011606_001 1.95	1	70	4,807.2	4,713.6
R011606_003 1.63	1	70	4,872.0	4,793.0
R021606_001 1.01	2	140	13,067.0	12,935.4
R041606_001 0.61	4	280	31,252.0	31,061.0
R061606_001 1.39	6	430	49,421.0	48,736.0
Average 1.32	—	—	—	—

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